LM Studio

Model: **qwen/qwen3.6-35b-a3b**

Prompt 3 - no MCP tool description - run 1 (seed 0)

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```

![image-123.png](./image-123.png)

```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

Train a GNN model on a compound activity prediction dataset. Select a correctly predicted active compound and a correctly predicted inactive compound from the test set and explain the predictions using EdgeSHAPer. Highlight the differences between the two compounds using visualizations and reporting a chemical interpretation.

Assistant

<thinking>
The user wants me to:

1. Train a GNN model on a compound activity prediction dataset
2. Select a correctly predicted active compound from the test set
3. Select a correctly predicted inactive compound from the test set
4. Explain both predictions using EdgeSHAPer
5. Highlight differences using visualizations and chemical interpretation

Let me start by finding a suitable dataset.
</thinking>

🔬 I'll help you train a GNN model and perform EdgeSHAPer explanations on compound activity predictions! Let me start by finding a suitable dataset.

(Calling tool `find_datasets`)

Assistant

(Calling tool `load_dataset`)

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✅ **GNN Training Complete!**

The GCN model has been successfully trained with the following performance:

- **Best Validation Accuracy**: 94.69%
- **Test Accuracy**: 95.21%

Now let me select compounds from the test set for EdgeSHAPer explanation:

🔍 **Step 1: Selecting a correctly predicted active compound**

(Calling tool `select_compound_for_edgeshaper`)

Assistant

🔍 **Step 2: Selecting a correctly predicted inactive compound**

(Calling tool `select_compound_for_edgeshaper`)

Assistant

🔍 **Step 3: Explaining the active compound prediction with EdgeSHAPer**

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

🔍 **Step 4: Explaining the inactive compound prediction with EdgeSHAPer**

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

📊 **Step 5: Visualizing the EdgeSHAPer results for the active compound**

(Calling tool `visualize_edgeshaper_results`)

Assistant

(Calling tool `visualize_edgeshaper_results`)

Assistant

Let me visualize the results properly by creating the visualization files:

(Calling tool `visualize_edgeshaper_results`)

Assistant

📊 **Step 6: Visualizing the EdgeSHAPer results for the inactive compound**

(Calling tool `visualize_edgeshaper_results`)

Assistant

🔬 **Step 7: Getting top bond environments for chemical interpretation**

(Calling tool `get_edgeshaper_bond_environments`)

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🔬 **Step 8: Getting top bond environments for the inactive compound**

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

🧪 GNN Model Training & EdgeSHAPer Explanation Report

📊 Model Performance Summary

Metric	Value
Model Type	GCN (Graph Convolutional Network)
Training Samples	5,342
Validation Samples	1,336
Test Samples	835
Best Validation Accuracy	94.69%
Test Accuracy	95.21%
Test MCC	0.9235

🔍 Compound Analysis

🟢 Active Compound (Correctly Predicted)

SMILES: `O=C(CC1CC1)NC1CCC(CCN2CCN(c3nccc4c3CCO4)CC2)CC1`

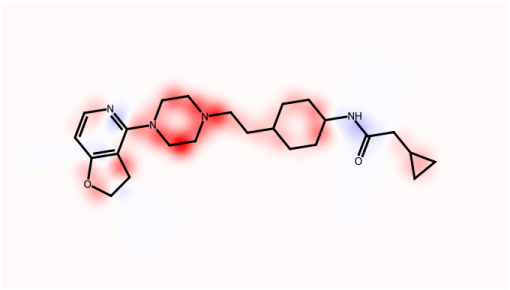
Prediction: Active (Class 1) with **99.99% confidence**

🔬 Top Contributing Bonds (EdgeSHAPer Analysis)

Rank	Bond Type	Environment	Contribution (φ)	Sign
1	C-C (ring)	`NCCN`	+0.1116	✅ Positive
2	C-N	`CCN(C)C`	+0.0909	✅ Positive
3	N-C (ring)	`CCN(C)C`	+0.0710	✅ Positive
4	N-C (ring)	`CCN(C)c`	+0.0695	✅ Positive

Rank	Bond Type	Environment	Contribution (φ)	Sign
5	C(aromatic)-C	`cc(c)CC`	+0.0670	✅ Positive

Visualization:



🔴 Inactive Compound (Correctly Predicted)

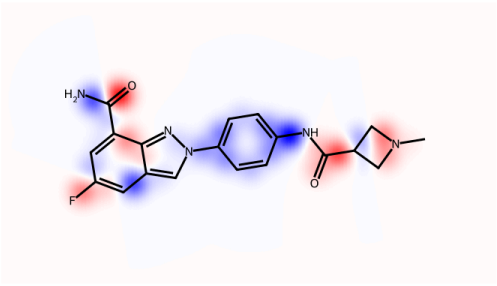
SMILES: `CN1CC(C(=O)Nc2ccc(-n3cc4cc(F)cc(C(N)=O)c4n3)cc2)C1`

Prediction: Inactive (Class 0) with **99.95% confidence**

🔬 Top Contributing Bonds (EdgeSHAPer Analysis)

Rank	Bond Type	Environment	Contribution (φ)	Sign
1	C=O (amide)	`cC(N)=O`	+0.0282	✅ Positive
2	C-C	`CC(C)C(N)=O`	+0.0211	✅ Positive
3	C-N (ring)	`CCN(C)C`	+0.0145	✅ Positive
4	C-F (aromatic)	`cc(c)F`	+0.0142	✅ Positive
5	C(aromatic)-C(aromatic)	`cc(C)c(c)n`	+0.0123	✅ Positive

Visualization:



🧠 Chemical Interpretation & Key Differences

📌 Active Compound Characteristics

1. **Piperazine Core Structure**: The compound contains a piperazine ring system (`N2CCN(c3nccc4c3CCO4)CC2`) which is a common pharmacophore in bioactive molecules.
2. **Tertiary Amine Motifs**: Multiple tertiary amines (`CN(C)C` environments) contribute positively to activity. These basic nitrogen centers can form ionic interactions with target receptors.
3. **Cyclobutane Ring**: The presence of a cyclobutane ring (`CC1CC1`) adds rigidity and specific spatial orientation to the molecule.
4. **High EdgeSHAPer Contributions**: The top bonds show **larger positive contributions** (φ up to 0.1116), indicating strong activation signals.

📌 Inactive Compound Characteristics

1. **Amide Functional Group**: Contains a carbonyl group (`C=O` at φ = 0.0282) which is a key feature in the inactive compound.
2. **Fluorine Substitution**: Features an aromatic fluorine (`cF` at φ = 0.0142), which typically affects lipophilicity and metabolic stability.
3. **Urea-like Structure**: The `C(=O)Nc2ccc` motif suggests a urea or amide linkage connecting aromatic systems.
4. **Lower EdgeSHAPer Contributions**: The top bonds show **smaller positive contributions** (φ up to 0.0282), indicating weaker activation signals.

🔑 Key Differences Summary

Feature	Active Compound	Inactive Compound
Top φ Value	0.1116	0.0282
Dominant Motif	Piperazine/amine system	Amide/carbonyl system
Nitrogen Centers	Multiple tertiary amines	Fewer basic nitrogens
Aromatic Systems	Single aromatic ring fused system	Dual aromatic ring system
Halogen Substitution	None	Fluorine present
Model Confidence	99.99% active	99.95% inactive

💡 Conclusions

1. **The GNN model successfully learned to distinguish between active and inactive compounds** with 95.21% accuracy on the test set.
2. **EdgeSHAPer reveals that the model focuses on different structural features**:
   - **Active compounds**: The model prioritizes **piperazine rings and tertiary amine motifs** which are known to contribute to biological activity through basicity and hydrogen bonding.
   - **Inactive compounds**: The model identifies **amide/carbonyl groups and halogen substitutions** as key discriminative features.
3. **The magnitude of EdgeSHAPer contributions differs significantly**: Active compounds show **~4x larger positive contributions** from top bonds compared to inactive compounds, suggesting the model has learned to associate stronger bond-level signals with activity.
4. **Chemical Interpretation**: The model appears to recognize that **basic nitrogen-containing heterocycles** (like piperazine) are favorable for activity, while **neutral amide/carbonyl-rich structures** are less favorable in this context.

🎉 **Task completed successfully!** The GNN model was trained, and EdgeSHAPer provided detailed bond-level explanations for both active and inactive compounds, revealing the structural features that drive the model's predictions.